

Generation Interconnection System Impact Study Report

For Queue Project Q13-0289

Gardner 230 kV
September 19, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0289. The Interconnection Customer proposes a 743.0 MW natural gas combined cycle facility interconnecting at the Gardner 230 kV substation in the Illinois area. The request was received on November 17, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 213 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 12 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$70,447,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The Gardner – Top Crop 230.0 kV is overloaded to 191.1% of its 454 MVA emergency rating for the loss of the Gardner – Grand Ridge Battery Projects 230.0 kV. The project contributes 100.0% DFAX (742.9 MW) to this overload.

Gardner – Grand Ridge Battery Projects 230.0 kV: 170.9% loading (508 MVA) following the outage of the Top Crop – Gardner 230.0 kV; DFAX = 100.0%.

For the loss of Muskingum Riv – Tidd Substation 500.0 kV, the Fishkill Plains – Dillard-Orange 345.0 kV line loads to 118.9% of its 1,399 MVA rating. Project DFAX: 6.4%.

For the loss of Ritter Park – Sugar Hill 345.0 kV L, the Medora – P.T.S 345.0 kV loads to 115.3% of its 1,258 MVA rating. Project DFAX: 8.7%.

Sugar Hill – Ritter Park Substation 345.0 kV: 115.2% loading (1,250 MVA) following the outage of the Hilltop – Piney Crk Project 500.0 kV; DFAX = 11.5%.

The Hampton Rd – Medora 345.0 kV is overloaded to 114.4% of its 1,243 MVA emergency rating for the loss of the Fishkill Plains Substation – Simpson Stuart Tap 345.0 kV. The project contributes 8.0% DFAX (59.3 MW) to this overload.

The Stewart Mountain Dam – Ritter Park Substation 345.0 kV Line is overloaded to 111.7% of its 1,173 MVA emergency rating for the loss of the Medora – P.T.S 345.0 kV. The project contributes 8.1% DFAX (59.9 MW) to this overload.

The Top Crop 500/230 kV Auto is overloaded to 110.4% of its 705 MVA emergency rating for the loss of the Grand Ridge Battery Projects – Gardner 230.0 kV. The project contributes 95.0% DFAX (706.2 MW) to this overload.

For the loss of Allegheny Energy Units 3 4 & 5 345-1 kV Auto, the Burwell – P.T.S 345.0 kV loads to 102.9% of its 1,075 MVA rating. Project DFAX: 8.1%.

For the loss of E Methuen 500-230 kV Transformer, the James De Young Substation – Carley Rd 230.0 kV loads to 102.7% of its 463 MVA rating. Project DFAX: 7.5%.

For the loss of Hilltop – Talenenergy Martins Crk Llc Williamsport 500.0 kV L, the Hilltop 500 kV/345 kV Transformer Bank loads to 102.5% of its 1,635 MVA rating. Project DFAX: 8.1%.

Montour Sta – Talenenergy Marins Crk Llc Williamsport Substation 345.0 kV: 100.1% loading (1,324 MVA) following the outage of the Orchard Rd – Ft Humbug 500.0 kV Line; DFAX = 5.9%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Gardner – Top Crop 230.0 kV	Rebuild line with higher-capacity conductor	\$7,093,000	86% (\$6,127,000)
Gardner – Grand Ridge Battery Projects 230.0 kV	Rebuild line with higher-capacity conductor	\$8,947,000	86% (\$7,729,000)
Dillard-Orange – Fishkill Plains 345.0 kV line	Rebuild line with higher-capacity conductor	\$87,044,000	10% (\$8,734,000)
Medora – P.T.S 345.0 kV	Rebuild line with higher-capacity conductor	\$29,345,000	10% (\$2,891,000)
Ritter Park – Sugar Hill 345.0 kV	Rebuild line with higher-capacity conductor	\$36,881,000	10% (\$3,622,000)
Hampton Rd – Medora 345.0 kV	Rebuild line with higher-capacity conductor	\$34,655,000	10% (\$3,415,000)
Stewart Mountain Dam – Ritter Park Substation 345.0 kV Line	Rebuild line with higher-capacity conductor	\$13,210,000	10% (\$1,301,000)
Top Crop 500/230 kV Auto	Replace transformer with higher-rated unit	\$33,490,000	86% (\$28,930,000)
Burwell – P.T.S 345.0 kV	Reconductor line	\$3,063,000	10% (\$301,000)
Carley Rd – James De Young 230.0 kV	Reconductor line	\$12,810,000	10% (\$1,223,000)
Hilltop 500 kV/345 kV XFMR	Replace transformer with higher-rated unit	\$27,989,000	10% (\$2,772,000)
Montour Sta – Talenenergy Marins Crk Llc Williamsport Substation 345.0 kV	Reconductor line	\$34,296,000	10% (\$3,402,000)
	Total allocated to project		\$70,447,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.